CIS 5500 Project Proposal
Group 13

1. A list of group members, group number, email addresses, and GitHub usernames
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2. A description of your application/website idea:

Our website will allow the user to explore housing and job market trends in various US geographies. It will display graphs and dashboards to the user which allow them to interactively explore the data.

3. For each dataset you've chosen:

ACS Census Data

a. This dataset comes from the American Community Society (ACS) 5-year estimates and reports the income related statistics for the US geographic areas. This dataset contains data regarding median incomes by city aggregated over a 12 month period. Each csv contains the data for one year, and data is available back to 2010.
The dataset contains estimates of median household income, number of households, and income distribution where the income is during the past 12 month period and values are adjusted for 2024 inflation adjusted dollars.

b. https://data.census.gov/table/ACSST5Y2024.S1901?q=S1901:+Income+in+the+Past+12+Months+(in+2024+Inflation-Adjusted+Dollars)&g=010XX00US$1600000

c. crawling NA

d. Stats:

i. relevant size statistics: ~30,000 rows per year. Each of the files are approximately 27-28 MB raw data and 6.7M cleaned. (~100MB).

ii. Summary stats: Minimum date from 2010 - Maximum date 2024.

5-year End Year	Mean Median Household Income ($)	Std. of Mediann HHI ($)	Mean # of Households	# of Places
2010	48,266	24,511	2,907	28,832
2011	49,205	24,769	2,919	28,855
2012	49,472	24,919	2,931	28,911
2013	49,593	24,887	2,943	28,645
2014	50,022	24,944	2,955	28,732
2015	50,873	25,000	2,974	27,630

2016	52,096	25,599	2,996	27,643
2017	54,083	26,514	3,024	27,501
2018	56,158	27,502	3,053	27,464
2019	58,238	28,536	3,077	27,372
2020	61,029	30,393	2,921	28,462
2021	64,397	31,783	2,969	28,269
2022	69,807	33,586	2,986	28,348
2023	72,866	34,593	3,016	28,372
2024	75,036	35,284	3,058	28,240

ZHVI dataset

a. Zillow Home Value Index (ZHVI): A measure of the typical home value and market changes across a given region and housing type. It reflects the typical value for homes in the 35th to 65th percentile range.

b. https://www.zillow.com/research/data/

c. crawling NA

d. Stats:

i. Relevant size stats: ~7M rows uncleaned, 4.7M cleaned (397 MB)

ii. Summary stats:

Median home value: $173075K,
Avg: $239269K,
stddev: $255107K
City Count: 21441
Min Date: 2000-01-31
Max Date: 2025-12-31

4. A list of at least 5 queries (in natural language) you could write for your datasets. Some of these should require complex SQL (aggregations, subqueries, joins, etc.)

Query 1 for ACS Census Data: Median household income summary stats

```
SELECT
    COUNT(*) AS n_regions,
    AVG(S1901_C01_012E::INTEGER) AS avg_median_income,
    MIN(S1901_C01_012E::INTEGER) AS min_median_income,
    MAX(S1901_C01_012E::INTEGER) AS max_median_income
FROM acs_income
WHERE S1901_C01_012E IS NOT NULL;
```

Query 2 for ACS Census Data: Counties in the top 25% of income

```
WITH income_ranked AS (
        SELECT
                geo_id,
                name AS county_name,
                S1901_C01_012E::INTEGER AS median_income,
                        NTILE(4) OVER (ORDER BY S1901_C01_012E::INTEGER) AS
income_quartile
        FROM acs_income
        WHERE S1901_C01_012E IS NOT NULL
)
SELECT
        county_name,
        Median_income
FROM income_ranked
WHERE income_quartile = 4
ORDER BY median_income DESC;
```

Query 3 for Zillow Data: Average Home Value by State

```
SELECT
   "StateName",
   AVG("2024-12-31") AS avg_home_value_2024
FROM zillow_housing
WHERE "RegionType" = 'County'
GROUP BY "StateName"
ORDER BY avg_home_value_2024 DESC;
```

Query 4 for Zillow Data: Top 3 most expensive counties per state

```
WITH ranked_counties AS (
   SELECT
      "StateName",
      "RegionName" AS county_name,
      "2024-12-31" AS home_value,
      RANK() OVER (
         PARTITION BY "StateName"
         ORDER BY "2024-12-31" DESC
      ) AS price_rank
   FROM zillow_housing
   WHERE "RegionType" = 'County'
)
SELECT
   "StateName",
   county_name,
   home_value
FROM ranked_counties
```

```
WHERE price_rank <= 3
ORDER BY "StateName", home_value DESC;
```

Query 5 for Join: Join census income + Zillow Home values, Compute housing affordability ratio, filter to high-income counties only

```
WITH high_income_counties AS (
   SELECT
      geo_id,
      name AS county_name,
      S1901_C01_012E::INTEGER AS median_income
   FROM acs_income
   WHERE S1901_C01_012E::INTEGER >
        (SELECT PERCENTILE_CONT(0.75)
         WITHIN GROUP (ORDER BY S1901_C01_012E::INTEGER)
         FROM acs_income)
)
SELECT
   h.county_name,
   z."StateName",
   h.median_income,
   z."2024-12-31" AS home_value_2024,
   ROUND(z."2024-12-31" / h.median_income, 2) AS price_to_income_ratio
FROM high_income_counties h
JOIN zillow_housing z
   ON LOWER(h.county_name) = LOWER(z."RegionName")
  AND z."RegionType" = 'County'
ORDER BY price_to_income_ratio DESC;
```